Lecture 27

Zero curvature and monodromy

The KdV Lax identity of Lecture 24 describes compatibility of linear operators. A related construction uses two first-order matrix equations. On a spatial circle their compatibility controls transport around one full period and gives conserved traces.

Objectives. *We prove the evolution law for periodic matrix transport with its basepoint term. We verify a matrix representation of sine-Gordon in physical coordinates and compute the transport of its vacuum.*

By the end of this lecture you will be able to:

- Derive monodromy evolution for a C^1 periodic flat matrix pair.
- Verify the converted sine-Gordon pair and the vacuum traces.
- Distinguish conserved traces from independence and Poisson commutativity.

27.1 Compatibility of the two linear equations

Fix an open interval $I \subset \mathbb{R}$, a period $L_{\mathrm{per}} > 0$ and a matrix size $n \geq 1$. For each fixed parameter λ in a set $\Lambda \subset \mathbb{C} \setminus \{0\}$, let $U(t,x;\lambda)$ and $V(t,x;\lambda)$ be complex $n \times n$ matrices, C^1 in $(t,x) \in I \times \mathbb{R}$ and periodic in x. The parameter is held constant when differentiating.

Definition 27.1 (Zero-curvature equation)**.** The pair is *flat*, or satisfies the *zero-curvature equation*, if

$$U_t - V_x + [U,V] = 0, \qquad [U,V] = UV - VU. \tag{27.1}$$

The associated linear equations for a column Ψ are $\Psi_x = U\Psi$ and $\Psi_t = V\Psi$. The variable λ is called a *spectral parameter* when such a family is used to represent the nonlinear field equation.

Indeed, formal differentiation gives $\Psi_{xt} - \Psi_{tx} = (U_t - V_x + [U,V])\Psi$. Thus (27.1) is the matrix compatibility condition, rather than an equality imposed only on a particular column. The connection convention of Lecture 19 is $A = -U\mathrm{d}x - V\mathrm{d}t$: its parallel equation $(\mathrm{d}+A)\Psi = 0$ has the stated signs. Direct expansion of $F = \mathrm{d}A + A \wedge A$ gives $F_{xt} = U_t - V_x + [U,V]$.

27.2 Transport and its time derivative

For a fixed λ, consider the initial matrix problem

$$Y_x(t,x) = U(t,x)Y(t,x), \qquad Y(t,0) = \mathbf{1}_n, \quad 0 \leq x \leq L_{\mathrm{per}}. \tag{27.2}$$

We suppress λ during the proof. Later spatial intervals multiply on the left, as Figure 27.1 shows. This order follows from $Y_x = UY$.

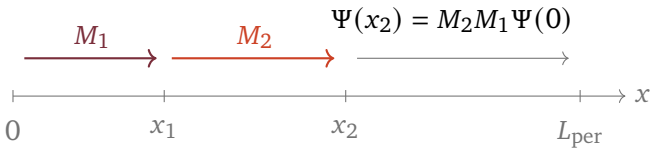


Figure 27.1: Transport first applies M_1 and then M_2 to a column. The later factor M_2 is therefore on the left.

Lemma 27.2 (Regularity of transport). *Problem* (27.2) *has a unique invertible solution. The matrices Y and Y_t are continuous, and*

$$(Y_t)_x = U_t Y + U Y_t, \qquad Y_t(t,0) = 0. \tag{27.3}$$

Proof. Iteration of $Y = \mathbf{1}_n + \int_0^x U(t,s)Y(t,s)\mathrm{d}s$ gives

$$Y = \mathbf{1}_n + \sum_{j\geq 1} \int_{0<s_j<\cdots<s_1<x} U(t,s_1)\cdots U(t,s_j)\,\mathrm{d}s_j \cdots \mathrm{d}s_1.$$

Use a submultiplicative matrix norm. On a compact time interval $K \subset I$, let $M \geq 1$ bound $\|U\|$, and let M_t bound $\|U_t\|$, for $0 \leq x \leq L_{\mathrm{per}}$. The jth term is bounded by $M^j L_{\mathrm{per}}^j / j!$. Its time derivative is bounded by $j M_t M^{j-1} L_{\mathrm{per}}^j / j!$. Both scalar series converge. Uniform convergence therefore permits termwise time differentiation. The integral equation gives the spatial derivative and, after time differentiation, (27.3).

For uniqueness, the difference of two solutions satisfies the homogeneous Volterra equation. Iterating j times bounds its supremum by its own finite supremum times $M^j L_{\mathrm{per}}^j / j!$. This factor tends to zero, so the difference vanishes. The same construction solves $Z_x = -ZU$, $Z(t,0) = \mathbf{1}_n$. Differentiating ZY gives zero, so $ZY = \mathbf{1}_n$. A square matrix with a left inverse is invertible. Hence $Z = Y^{-1}$. ∎

Definition 27.3 (Monodromy matrix). The *monodromy matrix* based at zero is

$$T(t;\lambda) = Y(t, L_{\mathrm{per}}; \lambda) = \mathcal{P} \exp \int_0^{L_{\mathrm{per}}} U(t,x;\lambda)\,\mathrm{d}x. \tag{27.4}$$

The symbol $\mathcal{P}$ denotes the ordered series just proved, with later x on the left. It does not mean the exponential of an ordinary integral when matrices at different positions do not commute.

A vector transported once around the spatial circle returns to the same basepoint with value $T\Psi(0)$. That value need not equal its initial value.

27.3 The monodromy evolves by a commutator

Theorem 27.4 (Monodromy evolution). *Under the stated C^1 and periodicity assumptions, flatness implies*

$$T_t = [V(t,0;\lambda), T]. \tag{27.5}$$

For every $\lambda \in \Lambda$ and positive integer r, the trace $\operatorname{tr}(T(t;\lambda)^r)$ is constant on I.

Proof. Define $W = Y_t - VY + YV(t,0)$. The transport lemma justifies its spatial derivative. Using flatness gives

$$\begin{aligned} W_x &= UY_t + (U_t - V_x - VU)Y + UYV(t,0) \\ &= U\big[Y_t - VY + YV(t,0)\big] = UW. \end{aligned}$$

Its value at zero is $0 - V(t,0) + V(t,0) = 0$. Uniqueness of the homogeneous matrix problem gives $W = 0$. At $x = L_{\mathrm{per}}$, this identity reads

$$T_t = V(t, L_{\mathrm{per}})T - TV(t,0). \tag{27.6}$$

Periodicity identifies the two endpoint values of V and proves (27.5). The right multiplication by $V(t,0)$ is required because the normalisation $Y(t,0) = \mathbf{1}_n$ has zero time derivative.

The product rule and cyclic invariance of finite matrix trace give

$$\frac{\mathrm{d}}{\mathrm{d}t}\operatorname{tr}(T^r) = r\operatorname{tr}(T^{r-1}T_t) = r\operatorname{tr}\big(T^{r-1}V(t,0)T - T^rV(t,0)\big) = 0.$$

This holds for each fixed parameter and every $t \in I$. ∎

No analytic dependence on λ was needed. The result concerns finite matrix transport with C^1 coefficients, so the unbounded-operator issue in Lecture 24 does not enter this proof.

27.4 The sine-Gordon representation in physical coordinates

A *classical sine-Gordon field* is a real C^2 function satisfying

$$\phi_{tt} - \phi_{xx} + \sin\phi = 0. \tag{27.7}$$

For the circle application, assume ϕ and its derivatives are periodic with period L_{per}. The statements below are conditional on such a field; no Cauchy theorem for this nonlinear equation is asserted here.

Recall the Pauli matrices of Lecture 14:

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

For $\lambda \in \mathbb{C} \setminus \{0\}$, use the light-cone coordinates $\xi = (x+t)/2$ and $\eta = (x-t)/2$. Then $\partial_\xi = \partial_x + \partial_t$ and $\partial_\eta = \partial_x - \partial_t$, so $\phi_{\xi\eta} = \phi_{xx} - \phi_{tt}$. The light-cone pair is

$$U_{\mathrm{lc}} = \frac{\mathrm{i}\lambda}{2}\sigma_3 + \frac{\mathrm{i}}{2}\phi_\xi\sigma_1, \qquad V_{\mathrm{lc}} = -\frac{\mathrm{i}}{2\lambda}(\cos\phi\,\sigma_3 + \sin\phi\,\sigma_2). \tag{27.8}$$

This is Dunajski, *Solitons, Instantons, and Twistors*, §3.3, equation (3.3.11), p. 49, with $\lambda = 2\lambda_D$. We state this light-cone pair and verify its converted pair directly. Since $\partial_x = (\partial_\xi + \partial_\eta)/2$ and $\partial_t = (\partial_\xi - \partial_\eta)/2$, that pair is

$$\begin{aligned} \phi_\xi &= \phi_x + \phi_t, \qquad N = \cos\phi\,\sigma_3 + \sin\phi\,\sigma_2, \\ U &= \frac{\mathrm{i}}{4}\big(\lambda\sigma_3 + \phi_\xi\sigma_1 - \lambda^{-1}N\big), \qquad V = \frac{\mathrm{i}}{4}\big(\lambda\sigma_3 + \phi_\xi\sigma_1 + \lambda^{-1}N\big). \end{aligned} \tag{27.9}$$

In particular, $U = (U_{\mathrm{lc}} + V_{\mathrm{lc}})/2$ and $V = (U_{\mathrm{lc}} - V_{\mathrm{lc}})/2$. Using the light-cone matrices unchanged as the (x,t) pair would be incorrect.

To verify (27.9), put $N_\phi = \cos\phi\,\sigma_2 - \sin\phi\,\sigma_3$. Then $N_t + N_x = \phi_\xi N_\phi$, and the derivative terms are

$$U_t - V_x = \frac{\mathrm{i}}{4}(\phi_{tt} - \phi_{xx})\sigma_1 - \frac{\mathrm{i}\phi_\xi}{4\lambda}N_\phi.$$

The Pauli commutators $[\sigma_3, \sigma_2] = -2\mathrm{i}\sigma_1$, $[\sigma_1, \sigma_3] = -2\mathrm{i}\sigma_2$ and $[\sigma_1, \sigma_2] = 2\mathrm{i}\sigma_3$ give

$$[U, V] = \frac{\mathrm{i}}{4}\sin\phi\,\sigma_1 + \frac{\mathrm{i}\phi_\xi}{4\lambda}N_\phi.$$

The terms involving N_ϕ cancel. Thus

$$U_t - V_x + [U, V] = \frac{\mathrm{i}}{4}(\phi_{tt} - \phi_{xx} + \sin\phi)\sigma_1. \tag{27.10}$$

This proves flatness if and only if the field satisfies sine-Gordon. For a periodic classical solution, U and V are C^1 and periodic. Theorem 27.4 therefore applies at every nonzero λ.

27.5 The vacuum on a circle

Example 27.5 (Vacuum monodromy)**.** For $\phi = 0$, both matrices are constant and diagonal:

$$U_0 = \frac{\mathrm{i}}{4}(\lambda - \lambda^{-1})\sigma_3, \qquad V_0 = \frac{\mathrm{i}}{4}(\lambda + \lambda^{-1})\sigma_3.$$

They commute. Solving $Y_x = U_0 Y$, $Y(0) = \mathbf{1}_2$, gives

$$T(\lambda) = \begin{pmatrix} e^{\mathrm{i}L_{\mathrm{per}}(\lambda-\lambda^{-1})/4} & 0 \\ 0 & e^{-\mathrm{i}L_{\mathrm{per}}(\lambda-\lambda^{-1})/4} \end{pmatrix}.$$

For every positive integer r its trace is

$$\operatorname{tr} T(\lambda)^r = 2\cos\left(\frac{rL_{\mathrm{per}}}{4}(\lambda - \lambda^{-1})\right).$$

These expressions are independent of time, as the theorem predicts. The determinant is one. At $\lambda = 1$, the matrix is the identity and every trace power is two. The parameter value $\lambda = 0$ remains excluded.

27.6 Conservation does not prove complete integrability

The theorem gives conserved functions labelled by λ and r. Different labels need not give independent functions of the field. For example, a family $F_\lambda = e^\lambda H$, for real λ, contains only multiples of one Hamiltonian, regardless of how many parameters are used.

Conserved quantities also need not have zero mutual Poisson brackets. For the canonical pairs (q_1, p_1) and (q_2, p_2) with Hamiltonian $H = p_1$, both q_2 and p_2 are conserved, while $\{q_2, p_2\} = 1$. A family is *in involution* when all its mutual brackets vanish. No such bracket calculation for the monodromy traces has been proved here. Nor have we proved that these quantities supply enough independent data to determine the general field evolution. Conservation, independence, involution and completeness are separate claims. A finite-dimensional Liouville–Arnold theorem alone does not establish any corresponding assertion for this field theory.

Summary. For a C^1 periodic flat matrix pair, the normalised spatial transport has monodromy $T_t = [V(t,0), T]$. Its trace powers are constant for every fixed spectral parameter. The converted sine-Gordon pair satisfies flatness exactly when $\phi_{tt} - \phi_{xx} + \sin\phi = 0$. The vacuum gives an explicit diagonal monodromy. These conserved traces do not by themselves establish independence, involution or completeness.

Self-check. A transport first acts by M_1 and then by M_2 on a column. Which product gives its final value, and which spatial ODE fixes that order?

Answer. The final value is $M_2 M_1 \Psi(0)$. The normalised transport solves $Y_x = UY$, so later spatial factors multiply on the left in its iterated integral series.

Self-check. Before using periodicity, what is T_t? Why can the term at $x = 0$ not be omitted?

Answer. It is $V(t, L_{\mathrm{per}})T - TV(t,0)$. The basepoint normalisation has $Y_t(t,0) = 0$. The term $-YV(t,0)$ in Y_t enforces that condition.

Self-check. Convert $\phi_{\xi\eta} = \sin\phi$ to (t,x) for $\xi = (x+t)/2$ and $\eta = (x-t)/2$.

Answer. The mixed derivative is $\phi_{xx} - \phi_{tt}$. Hence the equation becomes $\phi_{tt} - \phi_{xx} + \sin\phi = 0$. The two matrices must also be converted by their half sum and half difference.

Self-check. At the sine-Gordon vacuum, what are $T(1)$ and $\operatorname{tr} T(1)^r$? Does conservation for all λ prove independence of the charges?

Answer. They are $\mathbf{1}_2$ and 2 for every positive integer r. No. Parameter labels may give dependent functions, and conservation alone also does not establish their mutual Poisson commutativity.